University of Leeds



Group Kirlia

Coursework 1: Global Climate Attitudes and Eco-Anxiety Part I

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1 Introduction

Climate change constitutes not only a scientific and technological challenge but also a societal and political one. Public attitudes shape national policy priorities and influence the prospects for coordinated international action. The Peoples' Climate Vote 2024 dataset, compiled by the United Nations Development Programme in collaboration with the University of Oxford, provides nationally representative data from 73 countries—covering more than half of the global population—with demographic breakdowns by age, gender, and education. This report examines whether a consistent age gap exists in the perceived urgency of replacing coal, oil, and natural gas with renewable energy, with the aim of informing the design of targeted communication and policy interventions.

Existing research presents mixed, yet informative, evidence of generational differences in climate and energy attitudes. Hamilton, Hartter, and Bell (2019) report that younger adults in the United States express higher support for renewable energy development relative to older groups. Similarly, Roddis et al. (2018) find that younger respondents tend to favour a wider range of renewable technologies, whereas older adults show comparatively stronger support for nuclear energy and hydraulic fracturing. Official survey data echo these patterns: the UK Department for Energy Security and Net Zero's *Public Attitudes Tracker* (Department for Energy Security and Net Zero 2024) indicates that, while overall approval of renewable energy is high across all age groups, respondents aged 16–34 are more likely to express strong support than those aged 55 and above.

2 Data and Methodology

2.1 Data source

This analysis is based on the Peoples' Climate Vote 2024 dataset compiled by the United Nations Development Programme in partnership with the University of Oxford (Programme and Oxford 2024). The dataset includes nationally representative survey responses from 73 countries, covering over half of the world's population. It records public attitudes toward climate change, the energy transition and institutional trust, disaggregated by age, gender and education. For the purposes of this study, only country-level responses split by age group are used.

2.2 Variable construction

Of primary interest is the extent of *urgent support* for replacing fossil fuels with renewable energy. Reports in the data set include the percentage of respondents in each demographic group who chose "very urgent" or "urgent." To consider age comparisons, the discussion here focuses on two age categories that were continually available: 18–35 and 36–59. A country-level age gap is defined as:

$$\text{Age Gap} = \text{Support}_{18-35} - \text{Support}_{36-59},$$

expressed in percentage points. Positive values indicate stronger youth support, while negative values indicate that older adults express greater urgency.

2.3 Data preprocessing

To guarantee comparability across countries, all aggregate categories, such as "All Ages" and "All Education", were removed. Only age-specific responses within the "All Education" category remained. Countries lacking age-group responses were excluded from the related calculations.All support values were converted to numeric percentages, and country names were standardised to allow consistent indexing and plotting.

2.4 Analytical approach

The analysis proceeds in three stages. First, a heatmap visualises cross-country support patterns for both age groups, ordered by the magnitude of the age gap, to identify broad similarities and divergences. Second, horizontal bar charts highlight the countries with the largest positive and negative age gaps. Finally, a histogram with an overlaid kernel density estimate summarises the global distribution of age gaps, providing an aggregate view of whether differences cluster around zero. All stages are descriptive, as the dataset provides aggregated percentages rather than individual-level records, and no causal inference is attempted.

3 Exploratory data analysis

3.1 Cross-country support levels by age group

To obtain an overview of age-specific support patterns, Figure 1 plots urgent support for replacing fossil fuels with renewable energy among 18–35 and 36–59 year-olds for all countries in the dataset. Countries are ordered from left to right by the size of the age gap (younger minus older), so that those with stronger youth support appear on the left and those with smaller or reversed gaps appear on the right.

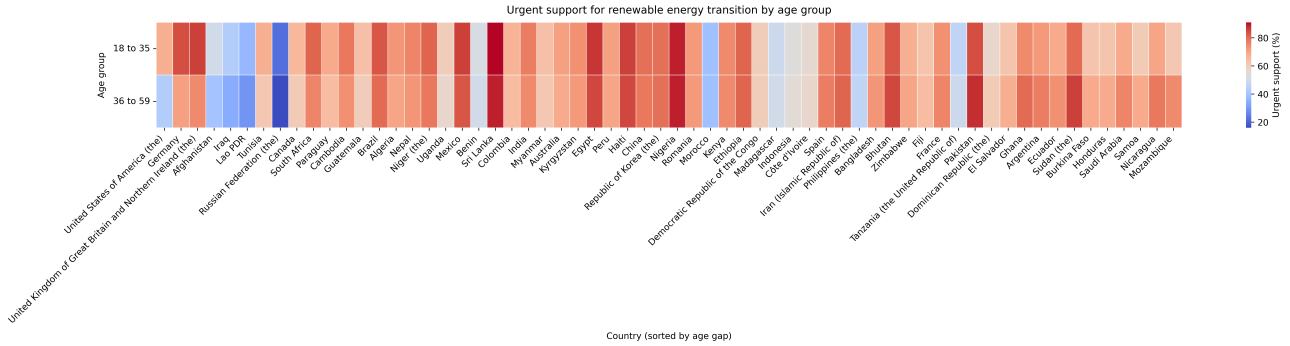


Figure 1: Urgent support for renewable energy transition by age group

Figure 1 shows that urgent support for the energy transition is generally high in both age groups, typically between about 60% and 80% in most countries. The colour differences between the two rows are modest for the majority of countries, indicating broadly similar levels of support among younger and older adults. Only a limited set of countries exhibits clear intergenerational divergence, which suggests that age alone is not a dominant determinant of support for the renewable energy transition at the global level.

3.2 Countries with the largest age gaps

While most countries show only small differences between age groups, it is still informative to identify where generational divides are most pronounced. Figure 2 therefore focuses on the ten countries with the largest positive age gaps (left panel) and the ten with the largest negative age gaps (right panel). Here, the age gap is defined as the percentage-point difference in urgent support between 18–35 and 36–59 year-olds.

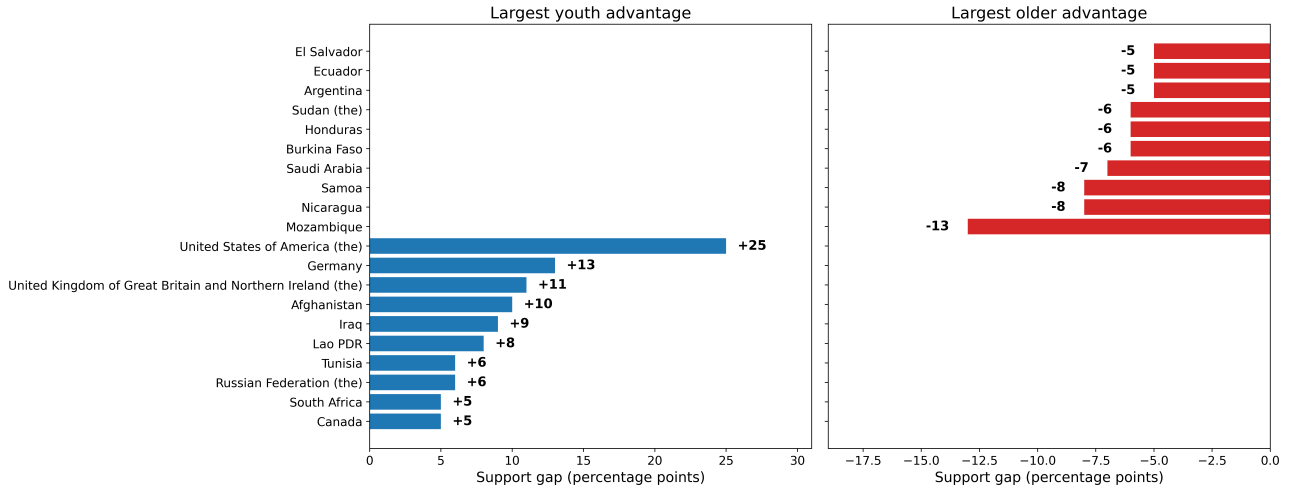


Figure 2: Countries with the largest youth and older-adult advantages in urgent support for renewable energy transition

In the left panel of Figure 2, the United States exhibits by far the largest youth advantage (+25 percentage points), followed by Germany (+13), the United Kingdom (+11), Afghanistan (+10) and Iraq (+9). In these countries, younger adults are substantially more likely than older adults to demand a rapid fossil-fuel phase-out. By contrast, the right panel highlights a smaller set of countries where older adults are more supportive than younger adults, most notably El Salvador (–13), Ecuador (–8) and Argentina (–8). These extremes illustrate that,

although the global average age gap is small, some national contexts display pronounced youth- or older-skewed patterns of support.

3.3 Global distribution of age gaps

Finally, to summarise the overall pattern of intergenerational differences, Figure 3 displays the distribution of age gaps across all countries. The histogram plots the number of countries in each gap interval, with an overlaid kernel density curve and vertical lines marking zero (no age difference) and the global mean age gap.

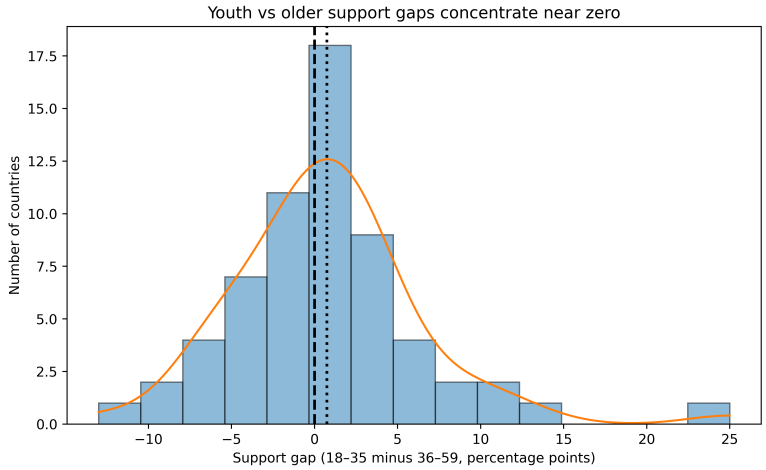


Figure 3: Distribution of age gaps in urgent support for the renewable energy transition

Figure 3 shows that age gaps are tightly concentrated around zero, with most countries lying between roughly –5 and +5 percentage points. A modest right tail corresponds to countries with strong youth advantages, while a thinner left tail represents a small number of older-advantaged cases. The mean age gap lies only slightly above zero, and the median is effectively zero, confirming that there is no consistent global tendency for younger or older adults to be systematically more supportive. Instead, generational differences in support for the energy transition are modest and highly context-dependent across countries.

4 Discussion

The exploratory analysis indicates that urgent support for the renewable energy transition is high across countries and broadly similar between younger (18–35) and older (36–59) adults. As shown in Figures 1 and 2, most countries display only small age differences, with sizeable youth or older-adult advantages appearing in only a few cases. The distribution in Figure 3 further confirms that age gaps cluster tightly around zero, suggesting no consistent generational divide at the global level.

Overall, these findings imply that age is not a major determinant of support for the energy transition. Where notable age differences do occur, they are likely shaped by country-specific political, social, or environmental contexts rather than by age alone. The analysis is limited by the use of aggregated data, incomplete coverage of the 60+ category, and the absence of controls for other demographic factors. Accordingly, the results should be interpreted as descriptive patterns rather than causal evidence.

5 Conclusion

This report set out to examine whether there is a consistent age gap across countries in public support for rapidly replacing coal, oil and gas with renewable energy. Using the Peoples' Climate Vote 2024 dataset, the exploratory analysis shows that although younger adults (18–35) are, on average, slightly more supportive than 36–59 year-olds, age gaps are small for most countries and their direction is heterogeneous. Some countries display pronounced youth advantages, while a smaller group shows older-advantaged patterns, but there is no global tendency for one age group to be systematically more supportive. Overall, the evidence suggests that public backing for the energy transition is broadly shared across generations, and that national context and other demographic or political factors are likely to be at least as important as age in shaping attitudes.

References

Department for Energy Security and Net Zero (2024). *Public Attitudes Tracker: Renewable Energy (Spring 2024)*. UK Government. Accessed 2024. URL: https://www.gov.uk/government/statistics/desnz-public-attitudes-tracker-spring-2024.

Hamilton, Lawrence C., Joel Hartter, and Erin Bell (2019). "Generation gaps in U.S. public opinion on renewable energy and climate change". In: *PLOS ONE* 14.7, e0217608. DOI: 10.1371/journal.pone.0217608.

Programme, United Nations Development and University of Oxford (2024). *Peoples' Climate Vote 2024: Global Report*. United Nations Development Programme. Dataset and documentation. URL: https://www.undp.org/publications/peoples-climate-vote-2024.

Roddis, Paul et al. (2018). "Accounting for the role of older age groups in renewable energy support". In: *Energy Research & Social Science* 35, pp. 11–22. DOI: 10.1016/j.erss.2018.02.023.